

WEEK 3: SELLER AND INVENTORY MANAGEMENT SYSTEM

Scenario

The pet e-commerce company sells pet food, toys, grooming products, accessories, and healthcare products supplied by multiple sellers. The system should maintain seller details, track products supplied by each seller, and monitor inventory levels to ensure that pet products are available for customers.

1. Create Seller and Inventory Tables

Seller Table

```
SQL> -- Seller Table
SQL> CREATE TABLE Seller (
  2     Seller_ID VARCHAR2(7),
  3     Seller_Name VARCHAR2(25),
  4     Email VARCHAR2(30),
  5     Phone NUMBER(10),
  6     Address VARCHAR2(50)
  7 );

Table created.
```

Inventory Table

```
SQL> -- Inventory Table
SQL> CREATE TABLE Inventory (
  2     Inventory_ID VARCHAR2(7),
  3     Product_ID VARCHAR2(7),
  4     Seller_ID VARCHAR2(7),
  5     Stock_Quantity NUMBER,
  6     Stock_Status VARCHAR2(11),
  7     Last_Updated DATE
  8 );

Table created.
```

2. Establish Relationships

Seller and Product

One seller can supply multiple pet products.

One pet product can be supplied by one or multiple sellers.

Seller —————< Product
One to Many

Product and Inventory

Each pet product has inventory information.

Inventory tracks the available stock of each product.

Product Inventory
 One to One

3. Primary Key and Foreign Key

PRIMARY KEY FOR BOTH THE TABLES:

Seller Table
seller_id → Primary Key

Inventory Table
inventory_id → Primary Key

```
SQL> ALTER TABLE Seller
  2  ADD CONSTRAINT PK_Seller PRIMARY KEY(Seller_ID);

Table altered.

SQL>
SQL> ALTER TABLE Inventory
  2  ADD CONSTRAINT PK_Inv PRIMARY KEY(Inventory_ID);

Table altered.
```

FOREIGN KEY FOR INVENTORY TABLE:

Inventory Table
product_id → Foreign Key
References Product(product_id)

```
SQL> ALTER TABLE Inventory
  2  ADD CONSTRAINT FK_Prod
  3  FOREIGN KEY(Product_ID)
  4  REFERENCES Product(Product_ID);

Table altered.

SQL>
SQL> ALTER TABLE Inventory
  2  ADD CONSTRAINT FK_Seller
  3  FOREIGN KEY(Seller_ID)
  4  REFERENCES Seller(Seller_ID);

Table altered.
```

4. Constraints

Make Seller Email Unique

```
SQL> ALTER TABLE Seller
  2  ADD CONSTRAINT Un_mail UNIQUE(Email);

Table altered.
```

Make Email NOT NULL

```
SQL> ALTER TABLE Seller
  2  MODIFY Email NOT NULL;

Table altered.
```

Check Stock Quantity is Positive

```
SQL> ALTER TABLE Inventory
  2  ADD CONSTRAINT CK_Quan CHECK(Stock_Quantity >= 0);

Table altered.
```

5. Insert Values into Seller Table

```
SQL> INSERT INTO Seller VALUES ('S001', 'Arun Pets', 'arunpets@gmail.com', 9876543210, 'Chennai');
1 row created.

SQL> INSERT INTO Seller VALUES ('S002', 'Happy Paws', 'happypaws@gmail.com', 9876543211, 'Chennai');
1 row created.

SQL> INSERT INTO Seller VALUES ('S003', 'Pet World', 'petworld@gmail.com', 9876543212, 'Vellore');
1 row created.

SQL> INSERT INTO Seller VALUES ('S004', 'Animal Care', 'animalcare@gmail.com', 9876543213, 'Madurai');
1 row created.

SQL> INSERT INTO Seller VALUES ('S005', 'Paw Palace', 'pawpalace@gmail.com', 9876543214, 'Coimbatore');
1 row created.

SQL> INSERT INTO Seller VALUES ('S006', 'Pet Hub', 'pethub@gmail.com', 9876543215, 'Salem');
1 row created.

SQL> INSERT INTO Seller VALUES ('S007', 'Furry Friends', 'furryfriends@gmail.com', 9876543216, 'Chennai');
1 row created.

SQL> INSERT INTO Seller VALUES ('S008', 'Pet Zone', 'petzone@gmail.com', 9876543217, 'Trichy');
1 row created.

SQL> INSERT INTO Seller VALUES ('S009', 'Paws Mart', 'pawsmart@gmail.com', 9876543218, 'Tirunelveli');
1 row created.

SQL> INSERT INTO Seller VALUES ('S010', 'Royal Pets', 'royalpets@gmail.com', 9876543219, 'Chennai');
1 row created.
```

```
SQL> select * from seller;
```

SELLER_	SELLER_NAME	EMAIL	PHONE
---------	-------------	-------	-------

ADDRESS

S001	Arun Pets Chennai	arunpets@gmail.com	9876543210
------	----------------------	--------------------	------------

S002	Happy Paws Chennai	happypaws@gmail.com	9876543211
------	-----------------------	---------------------	------------

S003	Pet World Vellore	petworld@gmail.com	9876543212
------	----------------------	--------------------	------------

SELLER_	SELLER_NAME	EMAIL	PHONE
---------	-------------	-------	-------

ADDRESS

S004	Animal Care Madurai	animalcare@gmail.com	9876543213
------	------------------------	----------------------	------------

S005	Paw Palace Coimbatore	pawpalace@gmail.com	9876543214
------	--------------------------	---------------------	------------

S006	Pet Hub Salem	pethub@gmail.com	9876543215
------	------------------	------------------	------------

SELLER_	SELLER_NAME	EMAIL	PHONE
---------	-------------	-------	-------

ADDRESS

S007	Furry Friends Chennai	furryfriends@gmail.com	9876543216
------	--------------------------	------------------------	------------

S008	Pet Zone Trichy	petzone@gmail.com	9876543217
------	--------------------	-------------------	------------

S009	Paws Mart Tirunelveli	pawsmart@gmail.com	9876543218
------	--------------------------	--------------------	------------

SELLER_	SELLER_NAME	EMAIL	PHONE
---------	-------------	-------	-------

ADDRESS

S010	Royal Pets Chennai	royalpets@gmail.com	9876543219
------	-----------------------	---------------------	------------

```
10 rows selected.
```

6. Insert Values into Inventory Table

```

SQL> INSERT INTO Inventory VALUES ('I001', 'P101', 'S001', 25, 'In Stock', DATE '2026-10-01');
1 row created.

SQL> INSERT INTO Inventory VALUES ('I002', 'P102', 'S002', 40, 'In Stock', DATE '2026-10-01');
1 row created.

SQL> INSERT INTO Inventory VALUES ('I003', 'P103', 'S003', 15, 'In Stock', DATE '2026-10-02');
1 row created.

SQL> INSERT INTO Inventory VALUES ('I004', 'P104', 'S004', 30, 'In Stock', DATE '2026-10-02');
1 row created.

SQL> INSERT INTO Inventory VALUES ('I005', 'P105', 'S005', 10, 'Low Stock', DATE '2026-10-03');
1 row created.

SQL> INSERT INTO Inventory VALUES ('I006', 'P106', 'S006', 50, 'In Stock', DATE '2026-10-03');
1 row created.

SQL> INSERT INTO Inventory VALUES ('I007', 'P107', 'S007', 8, 'Low Stock', DATE '2026-10-04');
1 row created.

SQL> INSERT INTO Inventory VALUES ('I008', 'P108', 'S008', 35, 'In Stock', DATE '2026-10-04');
1 row created.

SQL> INSERT INTO Inventory VALUES ('I009', 'P109', 'S009', 20, 'In Stock', DATE '2026-10-05');
1 row created.

SQL> INSERT INTO Inventory VALUES ('I010', 'P110', 'S010', 5, 'Low Stock', DATE '2026-10-05');
1 row created.

SQL>
SQL> COMMIT;

Commit complete.

```

INVENTO	PRODUCT	SELLER_	STOCK_QUANTITY	STOCK_STATU	LAST_UPDA
I001	P101	S001	25	In Stock	01-OCT-26
I002	P102	S002	40	In Stock	01-OCT-26
I003	P103	S003	15	In Stock	02-OCT-26
I004	P104	S004	30	In Stock	02-OCT-26
I005	P105	S005	10	Low Stock	03-OCT-26
I006	P106	S006	50	In Stock	03-OCT-26
I007	P107	S007	8	Low Stock	04-OCT-26
I008	P108	S008	35	In Stock	04-OCT-26
I009	P109	S009	20	In Stock	05-OCT-26
I010	P110	S010	5	Low Stock	05-OCT-26

10 rows selected.

7. Report 1: Seller-wise Product Report

Display: Seller name, Number of products supplied, Available stock

```
SQL> SELECT s.Seller_Name,
2         COUNT(i.Product_ID) AS Number_of_Products,
3         SUM(i.Stock_Quantity) AS Available_Stock
4 FROM Seller s
5 JOIN Inventory i
6 ON s.Seller_ID = i.Seller_ID
7 WHERE i.Stock_Status = 'In Stock'
8 GROUP BY s.Seller_Name;
```

SELLER_NAME	NUMBER_OF_PRODUCTS	AVAILABLE_STOCK
Happy Paws	1	40
Animal Care	1	30
Pet World	1	15
Pet Zone	1	35
Paws Mart	1	20
Arun Pets	1	25
Pet Hub	1	50

7 rows selected.

8. Report 2: Stock Availability Report

Display: Product name, Stock quantity, Stock status

```
SQL> SELECT p.Product_Name,
2         i.Stock_Quantity,
3         i.Stock_Status
4 FROM Product p
5 JOIN Inventory i
6 ON p.Product_ID = i.Product_ID;
```

```
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATU
-----
pedigree dog food
                25 In Stock

cat dry food
               40 In Stock

bird seed mix
               15 In Stock
```

```
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATU
-----
fish food flakes
               30 In Stock

puppy chew toy
               10 Low Stock

cat ball toy
               50 In Stock
```

```
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATU
-----
pet shampoo
               8 Low Stock

Dog collar
               35 In Stock

pet vitamin supplement
               20 In Stock
```

```
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATU
-----
pet sleeping bed
               5 Low Stock
```

10 rows selected.

9. Report 3: Inventory Analysis

Find: Total products available, Products out of stock, Highest stocked products, Average inventory quantity

Total Products Available

```
SQL> SELECT COUNT(*) AS Total_Available_Products
2 FROM Inventory
3 WHERE Stock_Status = 'In Stock';

TOTAL_AVAILABLE_PRODUCTS
-----
7
```

Products Out of Stock

```
SQL> SELECT COUNT(*) AS Low_Stock_Products
2 FROM Inventory
3 WHERE Stock_Status = 'Low Stock';

LOW_STOCK_PRODUCTS
-----
3
```

Highest Stocked Products

```
SQL> SELECT p.Product_Name, i.Stock_Quantity
2 FROM Product p
3 JOIN Inventory i
4 ON p.Product_ID = i.Product_ID
5 WHERE i.Stock_Quantity = (
6     SELECT MAX(Stock_Quantity)
7     FROM Inventory
8 );

PRODUCT_NAME
-----
cat ball toy
50
```

Average Inventory Quantity

```
SQL> SELECT AVG(Stock_Quantity) AS Average_Inventory_Quantity
2 FROM Inventory;

AVERAGE_INVENTORY_QUANTITY
-----
23.8
```